

DDoS Incident Analysis —Case Studies

1) Cloudflare — 7.3 Tbps DDoS Attack (May 2025)

Target:

This attack targeted a single IP address protected by Cloudflare. The exact victim was not publicly disclosed, but Cloudflare stated it was a high-value customer likely operating critical online services.

Technology Used:

The adversaries launched a hyper-volumetric UDP flood attack. The botnet generated massive bursts of spoofed packets leveraging reflection and amplification from vulnerable public services. The attack peaked at 7.3 Tbps and delivered approximately 37.4 TB of data in about 45 seconds. Its short-lived yet intense nature suggests the use of modern “burst-style” botnets designed to evade long-term analysis.

Attacker Motive:

Investigators suspect capability demonstration and defensive reconnaissance. These types of bursts are commonly used to probe response speed and identify weak routing paths before extortion or subsequent persistent attacks.

Impact:

Cloudflare automatically detected and mitigated the wave using its Anycast global network. Regular internet users did not experience major outages, but a small organization without similar defenses would likely suffer complete service disruption, routing failures, and possible collateral ISP congestion.

Defensive Strategies:

- Provision global scrubbing capacity with Anycast load distribution
- Implement per-protocol rate limiting and anomaly-based traffic filtering
- Avoid exposing a single point-of-failure IP to the public network
- Conduct DDoS response exercises & include escalation playbooks
- Deploy constant monitoring to track short-duration bursts

2) Cloudflare — 11.5 Tbps DDoS Attack (September 2025)

Target:

A Cloudflare customer operating services dependent on continuous availability. Exact identity concealed due to security concerns.

Technology Used:

A sophisticated multi-vector flood combining high bandwidth throughput and extreme packets-per-second pressure. It weaponized thousands of distributed endpoints capable of packet manipulation and reflection. The peak of 11.5 Tbps broke previous global records, signifying the rapid growth of botnet capacity through IoT and exploited cloud infrastructure.

Attacker Motive:

Likely extortion or strategic disruption. Criminal actors commonly send large bursts as warnings preceding ransom messages. The sophistication indicates experienced threat actors with a goal of benchmarking Cloudflare's defense thresholds.

Impact:

Despite the magnitude, Cloudflare's automation absorbed the attack. If executed against smaller ISPs or on-prem infrastructures, this could have produced hours of downtime, routing blackholes, and even upstream carrier instability.

Defensive Strategies:

- Partner with enterprise-grade scrubbing providers operating at Tbps scale
- Deploy automated signatures + machine learning based anomaly detection
- Introduce emergency traffic sinkholing or geofencing to isolate malicious sources
- Use load balancing and redundant network entry points to absorb surges

3) Gcore — 6 Tbps Attack (October 2025)

Target:

Gcore, a global infrastructure provider servicing game publishers. Disruption led to lag and intermittent outages for connected game regions.

Technology Used:

UDP floods leveraging a distributed botnet, suspected to be AISURU, known for abusing unsecured SOHO routers and cameras. Packets-per-second hit approximately 5.3 billion pps. Attackers deliberately targeted latency-sensitive gaming traffic, where even seconds of disruption affect the user experience.

Attacker Motive:

Gaming services are frequent targets of rival groups, extortion gangs, and dissatisfied player communities. The absence of ransom claim in public disclosures suggests either competitive sabotage or attack rehearsal.

Impact:

Due to pre-established protections, Gcore prevented major downtime. However, even brief latency spikes harmed gameplay responsiveness, frustrating users and reducing trust in hosted platforms.

Defensive Strategies:

- Use Anycast routing with regional load shifting for real-time defense
- Deploy UDP-specific throttling, connection token validation, and game traffic prioritization
- Maintain global alerting and automated routing overrides during gaming peak hours
- Ensure peering agreements include emergency packet scrubbing

4) Microsoft Azure — DDoS-linked Global Outages (Mid-2024)

Target:

Microsoft's Azure cloud services, affecting government agencies, enterprises, and millions of daily users globally whose apps depend on Microsoft infrastructure.

Technology Used:

A mixed campaign of volumetric layer-3 floods and targeted layer-7 HTTP request spikes. At certain points, defensive configuration errors during mitigation cascaded into prolonged service degradation. Attackers exploited the complexity of global cloud traffic engineering.

Attacker Motive:

Reports pointed toward politically motivated hacktivist groups attempting to disrupt Western critical infrastructure. Banks, airlines, and public service portals indirectly suffered.

Impact:

Major slowdowns for SharePoint, Microsoft Teams, Exchange Online, and Azure-hosted websites. Public confidence and business operations were temporarily undermined, leading to financial impact for dependent sectors.

Defensive Strategies:

- Conduct disaster scenario testing for cloud-scale mitigation policies
- Enable autoscaling for protective services without manual misconfiguration
- Enhance routing resilience with backup scrubbing providers
- Maintain rapid user communication channels for transparency during incidents

5) Financial Sector Multi-Target DDoS Campaigns (2024–2025)

Target:

Banks and financial service organizations worldwide, especially online banking portals and transactional APIs relied upon for funds access.

Technology Used:

Attackers used a hybrid strategy: bandwidth floods to disrupt the main site + targeted HTTP floods to overload login and payment APIs. These attacks adapted to bypass traditional defenses by mimicking legitimate customer traffic.

Attacker Motive:

Strong financial motives including ransom demands, account takeover cover operations, or deliberate destabilization during high-volume trading hours. Some campaigns align with geopolitical tensions and state-sponsored objectives.

Impact:

Blocked customer logins, delayed payments, and reputational harm. Institutions also suffered increased operational costs due to emergency mitigation measures and regulatory scrutiny following outages.

Defensive Strategies:

- API-focused defenses like strict authentication, behavioral analysis, and micro-rate limiting
- Collaboration through FS-ISAC threat intelligence sharing

- Multi-cloud routing readiness to avoid regional chokepoints
- Continuous traffic analysis to detect fraud hidden in noise of DDoS spikes